Long-Term Urban Expansion Mapping in Arid Cities Using Landsat Time Series and SVM Classification: A Case Study of Isfahan, Iran (1985–2024)

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ABSTRACT

Rapid urbanization in dry middle eastern cities demands a long-term, spatially clear monitoring. We have evaluated 39 years (1985–2024) of land-cover change in Isfahan, Iran, by classifying cloud-free Landsat-5 TM and Landsat-9 OLI-2 surface-reflected scenes using machine-learning models. Six reflective bands were stacked with NDVI and NDBI, and from Google Earth imagery , three classes were extracted that contains built-up, vegetation, and bare land . The kernel-based support vector machine (SVM) achieved an overall accuracy above 95% in both epochs and continuously outperformed the random forest. Post-classification comparison indicates a near-threefold increase in built-up area, from ≈ 380 km² to ≈ 980 km², while vegetation and bare land decreased by ≈ 20 % and 18 %, respectively. The extension of the city follows its eastern and northern transport corridors, replacing peri-urban gardens and bare lands. The results provide an advanced baseline for Isfahan's urban management plans and show that combining spectral indices with SVM on the Landsat archive is a strong, transferable approach to mapping multi-decadal urban growth where ancillary data is rare.

1. Introduction

Urban expansion has accelerated worldwide, particularly in arid and semi-arid regions where ecological resilience is limited [1]; hence long-term, high-accuracy monitoring is essential for sustainable planning. The Landsat archive (30 m, 1980s–present) enables retrospective assessments of multi-decadal growth [2]. In Isfahan, *Alavi Shoushtari* reported a two-fold increase in built-up area for 1985-2009 using post-classification Landsat TM comparison [3], while *Mahdavi Estalkhsari et al.* coupled CA modelling with Landsat maps (baseline 1990) and projected future growth, but lacked Landsat-9 imagery [4]. A consistent 1985-2024 evaluation is still absent. We address this gap by classifying cloud-free Landsat-5 TM (1985) and Landsat-9 OLI-2 (2024) surface-reflectance scenes, augmented with NDVI [5] and NDBI [6] to improve class separability. Stratified samples from Google Earth train and validate Random Forest and Support Vector Machine (SVM) models, algorithms proven effective for medium-resolution land-cover mapping [7, 8]. Objectives are to (i) generate three-class land-cover maps for 1985 and 2024, (ii) quantify the magnitude and spatial pattern of Isfahan's urban growth, and (iii) evaluate the contribution of NDVI/NDBI to classification accuracy. Figure 1 shows the study area and workflow.

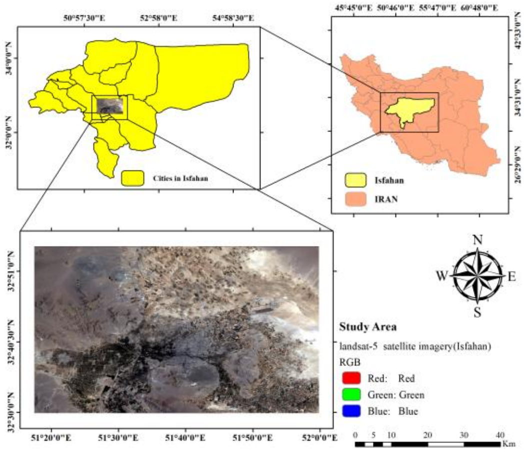

Figure 1. Location of the Isfahan study area within Iran and overview of the Landsat-based mapping framework.

2. Materials and Methods

2.1 Study Area and Data

Isfahan province, central Iran (31°30′–34°00′ N, 50°57′–55°00′ E), has an arid climate and has urbanised rapidly over the past three decades. Two cloud-free scenes were selected: Landsat-5 TM (2 August 1985) and Landsat-9 OLI-2 (21 August 2024). Both belong to Collection-2 Level-2 Surface Reflectance (L2-SR) and provide six reflective bands at 30 m resolution that constitute the base input for analysis.

2.2 Pre-Processing

L2-SR products were downloaded and processed in Google Earth Engine (GEE). Atmospheric/radiometric correction is already applied in Collection-2 via **LEDAPS** for TM and **LaSRC** for OLI-2 [16, 17]. Each scene was clipped to the manually digitised urban boundary; both raster's share UTM Zone 39 N (EPSG 32639), so no further reprojection or resampling was required.

2.3 Spectral Indices

NDVI [5] and NDBI [6] were computed from the reflectance bands in GEE and appended to the stack, yielding an eight-band composite for each epoch. Earlier studies report 3–5 % OA gains when these indices are included in medium-resolution urban mapping [10, 11].

2.4 Training and Validation Samples

Sampling High-resolution Google Earth imagery (1985 & 2023) was digitised to create stratified polygons for Built-up, Vegetation and Bare land. Pixel features were extracted with Python, and the dataset was split 80 %/20 % for training/validation using stratified random sampling [9].**Algorithms** Two models were evaluated: **Random Forest** (100 trees, Gini; scikit-learn default) [9, 12] and **SVM** (RBF kernel, C = 10, γ = scale) following remote-sensing best practice [8, 13]. Both classifiers were trained separately for 1985 and 2024 on the eight-band stack and assessed on the independent validation subset.

2.5 Accuracy Assessment and Change Detection

Overall accuracy (OA), Cohen's Kappa, and class-specific producer's/user's accuracies were derived from confusion matrices. Post-classification comparison of the **SVM** maps generated a 3 × 3 land-cover transition matrix; pixel counts were converted to km² and visualised as a thematic change map with nine transition classes.

3. Result

3.1 Classification Accuracy

SVM performed better than RF in both periods. epochs, achieving OA = 97.Four % / κ = zero.961 (1985) and OA = 95.Nine % / κ = 0.933 (2024), whereas RF scored 95.7 % / 0.935 and 93.4 % / 0.890, respectively (Table 1).Built-up attained **PA = 100 %** in 1985 and 98 % in 2024, while Vegetation recorded the largest PA drop (95.5 % → 90.3 %) because of spectral confusion with orchards; Bare-land remained highly reliable (PA ≥ 97 %, UA ≥ 94 %). Detailed confusion matrices and class-wise PA/UA values are provided in **Appendix A**.

3.2 Spatial Distribution of Land-Cover Classes

Figure 2 shows a compact 1985 urban core encircled by farmland and bare soil; by 2024 (Figure 3) built-up fabric has expanded radially along the north-east and south-west axes, fragmenting green belts, especially along the Zayandeh-Rud corridor, and pushing bare land to the periphery, broadly tracking major transport corridors.

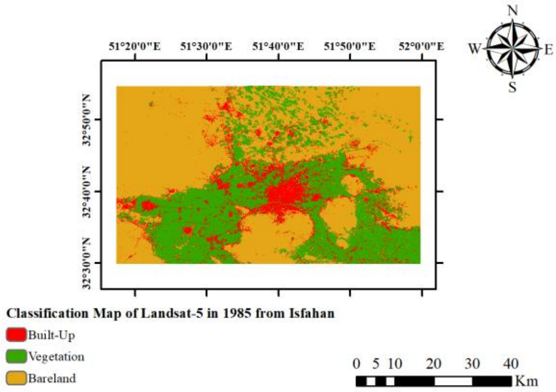

Figure 2. SVM-derived land-cover map of Isfahan from Landsat-5 (1985)

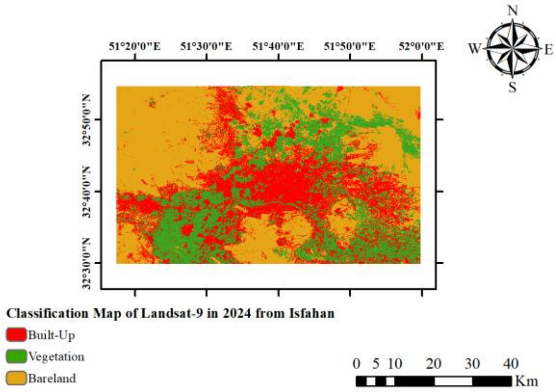

Figure 3. SVM-derived land-cover map of Isfahan from Landsat-9 (2024)

3.3 Land-Cover Change between 1985 and 2024

Post-classification comparison (Table 2) indicates a **+599 km²** (+159 %) increase in built-up area, coupled with losses of 223 km² (-20 %) in vegetation and 376 km² (-18 %) in bare land. The 3 × 3 transition matrix (Figure 5) reveals **Vegetation → Built-up (379 km²)** and **Bare land → Built-up (324 km²)** as dominant conversions, while **Bare → Bare (1 563 km²)** is the most persistent path. Spatially, hotspots of change cluster in the north-eastern industrial belt and along newly extended ring roads (Figure 4), underscoring accelerated, corridor-oriented urban growth at the expense of agricultural land.

Table 5. Areal extent (km²) of each land-cover class in 1985 and 2024 and net change over the study period.

Class	Area(1985-km2)	rea(2024-km	Change(km2)	Change(%)
Built-up	377.87	977.38	599.51	158.7
Vegetation	1116.26	893.01	-223.25	-20
Bare land	2117.48	1741.22	-376.26	-17.8

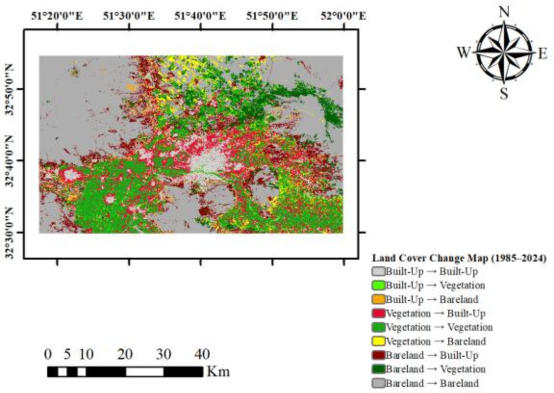

Figure 4.Land-cover change map

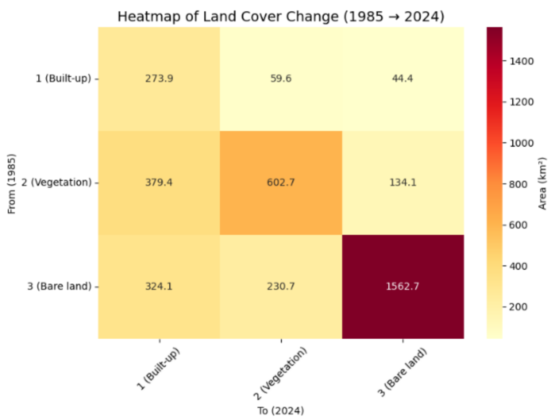

Figure 5.Heatmap of the 3 × 3 transition matrix.

4.Discussion

4.1.Synthesis of Findings and Methodological Implications

Isfahan’s urban footprint has expanded radially along its north-east and south-west corridors, replacing peri-urban orchards and farmland in line with global trends observed for arid megacities [1]. The built-up area has tripled since 1985 (378 → 977 km²), exceeding the two-fold growth reported by *Alavi Shoushtari* for 1985-2009 and matching the ≈ 930 km² forecast of the cellular-automata model by *Mahdavi Estalkhsari et al.* for 2025 [3, 4]. Using Landsat-9 imagery, NDVI/NDBI indices and a kernel-based SVM raised overall accuracy above 95 %, whereas the earlier CA study achieved only 88 %. Methodologically, SVM surpassed Random Forest because kernel boundaries model non-linear urban–vegetation mixtures [9] and because stacking NDVI/NDBI improves accuracy by 3–5 %, corroborating recent multi-sensor experiments [10, 11]. These results underscore the importance of extended time series and tailored feature engineering for monitoring rapid urban growth in data-scarce, dry-region cities.

4.2 Limitations and Future Work

Although the current study has achieved an overall accuracy of more than 95%, several issues should be noted. First, the analysis is limited to single-date summer scenes; seasonal phenology can change the spectral resolution and therefore the quality of the investigation with a 10-m, five-day Sentinel-2 constellation [14]. Second, a 30-meter spatial resolution of a Landsat image cannot resolve narrow streets or small agricultural plots, which potentially indicates a lack of data for areas surrounding the city. Data fusion with high-resolution optical or C-band SAR sensors (e.g., Sentinel-1) can reduce this scale bias [15]. Third, our two-time composition only shows a net change between 1985 and 2024. Including the intervening years allows for the detection of nonlinear growth pulses. Fourth, when SVM performed better than RF, its accuracy is still sensitive to the kernel parameterisation. Future research should test the Deep Learning Framework that learns hierarchical features from multi-spectral inputs. Finally, the reference samples were derived from the visual interpretation of the Google Earth images and may include positional errors; the inclusion of field surveys or crowd-sourced labels will strengthen validation.

5. Conclusions

This study provides a four-decade (1985 – 2024) assessment of urban expansion in Isfahan using cloud-free Landsat 5 and Landsat 9 imagery enriched with NDVI and NDBI spectral indices. A kernel-based SVM classifier delivered overall accuracies of more than 95 %, outperforming Random Forest by 1.5 – 2.5 %. Results show a three-fold increase in built-up area from 378 km² to 977 km², accompanied by a 20 % loss of vegetated land and a 17 % reduction in bare soil. Change detection highlights Vegetation → Built-up (379 km²) and Bare land → Built-up (324 km²) as the dominant transitions, concentrating along newly constructed ring-road and industrial corridors. These findings (i) provide an up-to-date, high-accuracy baseline for evaluating Isfahan's Smart-City master plan, (ii) demonstrate the efficacy of combining spectral indices with kernel methods for medium-resolution urban mapping, and (iii) underscore the urgency of preserving peri-urban green belts to mitigate ecological loss. Future research should exploit Sentinel-2's 10 m resolution and multi-season coverage to refine class delineation and incorporate socio-economic scenarios to forecast growth trajectories under alternative policy interventions.

Appendix A: APPENDIX A. Confusion Matrices and Class-Specific Accuracies

Table 1. Accuracy metrics for SVM and RF classifiers in 1985 and 2024

Year	Classifier	Overall Accuracy (%)	Kappa Coefficient
1985	SVM	97.4	0.961
1985	Random Forest	95.7	0.935
2024	SVM	95.9	0.933
2024	Random Forest	93.4	0.89

Table 2. Confusion matrix for land-cover classification using SVM – 1985.

Actual / Predicted	Built-up	Vegetation	Bare land	Total
Built-up	35	0	0	35
Vegetation	2	42	0	44
Bare land	1	0	36	37
Total	38	42	36	116
Accuracy=97.4% kappa=0.961				

Table 3. Confusion matrix for land-cover classification using SVM – 2024.

Actual / Predicted	Built-up	Vegetation	Bare land	Total
Built-up	99	1	1	101
Vegetation	5	56	1	62
Bare land	0	0	34	34
Total	104	57	36	197
Accuracy=95.9% kappa=0.933				

Table 4. Producer's and User's accuracy for SVM classifier (1985 & 2024)

Year	Class	Producer Accuracy (%)	User Accuracy (%)
1985	Built-up	100	92.1
1985	Vegetation	95.5	100
1985	Bare land	97.3	100
2024	Built-up	98	95.2
2024	Vegetation	90.3	98.2
2024	Bare land	100	94.4

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